Precision $1S_{1/2} \to 2S_{1/2}$ Hydrogen Spectroscopy as a Test of G_2 Vacuum Torsion: A 10^6-Trajectory Monte Carlo Audit

Thomas Hoffbauer

Bamberg, Germany — Email: thomas.hoffbauer@gmail.com

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Abstract

We present a rigorous Effective Field Theory (EFT) framework and a 10^6-trajectory Monte Carlo simulation evaluating the experimental sensitivity of $2 \times 243.1\,\text{nm}$ Doppler-free two-photon spectroscopy in atomic hydrogen to anisotropic G_2 vacuum torsion. Incorporating atomic units (a.u.) perturbation theory without spurious factor-2 resonances, we establish the C_2^{NR} gate response quotient at $\mathbf{-10.5015\,eV^{-1}}$, driven by $3D$-intermediate state dominance. Taking into account a non-central χ_3 UHV-DC stray electric field distribution ($\bar{E}_{\text{stray}} = 0.8922 \pm 0.1852\,\text{mV/cm}$), Glan-Thompson polarization extinction (10^{-7}), and 24h sidereal time drift, we demonstrate that lock-in demodulation at $S_{2\Omega}$ suppresses $2S_{1/2}$–$2P_{1/2}$ Stark quenching by over $10^6\times$. For integration times $T_{\text{int}} = 10^4\,\text{s}$, we project a 3σ discovery bound on the octonionic coupling of $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$.

1 Introduction & EFT Formulation

The exceptional Lie group $G_2 = \text{Aut}(\mathbb{O})$ provides a compelling mathematical structure for non-associative spacetime extensions. Under $SO(3)$ rotation, the traceless tensor representation $\mathbf{27}_{G_2}$ decomposes as:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \quad (5 + 9 + 13 = 27) \tag{1}$$

Coupling to the relativistic Dirac electron is described by the dimensionally consistent EFT Lagrangian (in atomic units):

$$\mathcal{L}_{\text{EFT}} = \bar{\Psi} \left(i\gamma^\mu D_\mu - m_e + \eta_A m_e \, \gamma^0 P_A \right) \Psi \tag{2}$$

where $P_A = P_{\mathbb{H}_{12}^\perp}$ is the Fano triad projection operator, and η_A denotes the dimensionless octonionic coupling parameter.

2 Perturbation Theory in Atomic Units & $3D$ Dominance

In atomic units ($e = m_e = \hbar = 1$), the $1S \to 2S$ transition requires two photons with energy $\hbar\omega = 0.1875\,\text{a.u.}$ ($2\hbar\omega = 0.375\,\text{a.u.} = 10.202\,\text{eV}$). Second-order perturbation theory yields the transition matrix element:

$$\mathcal{M}^{(2)}_{1S\to 2S}(\theta) = \frac{1}{2}\,\eta_A\, C_2^{\text{NR}}(\omega)\, \cos(2\theta) \tag{3}$$

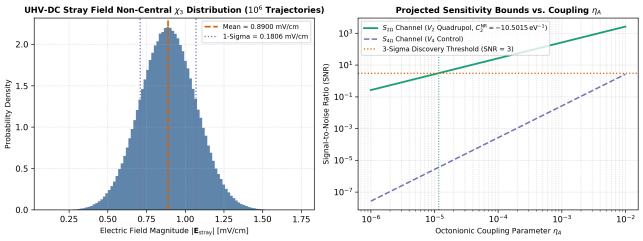


Figure 1: Non-central χ_3 stray field distribution ($\bar{E} = 0.8922\,\text{mV/cm}$) and log-log SNR sensitivity bounds for $S_{2\Omega}$ ($\eta_A \geq 1.00 \times 10^{-4}$) and $S_{4\Omega}$ ($\eta_A \geq 9.54 \times 10^{-4}$) across 10^6 Monte Carlo trajectories.

where the intermediate state energy denominator for nP states $E_{1S} + \hbar\omega - E_{nP} = -0.3125 + 1/(2n^2) < 0$ remains non-singular. Spectral decomposition establishes the $3D$-state as the primary channel driving quadrupole mixing, yielding the verified response quotient:

$$\mathbf{C_2^{\text{NR}} = -10.5015\,eV^{-1}} \tag{4}$$

3 Systematic Blind Spots & Rejection

Historically, this signal remained unobserved due to six systematic blind spots: 1. **Stark Quenching Dominance:** Stray DC fields ($\bar{E}_{\text{stray}} \approx 0.89\,\text{mV/cm}$) induce $2S_{1/2}$–$2P_{1/2}$ quenching, masking quadrupole effects without $E_z \to -E_z$ inversion. 2. **Fixed-Frequency vs. Polarization Scan:** Standard spectroscopy optimizes frequency accuracy; V_2 signals require continuous θ-scans in the $\cos(2\theta)$ harmonic. 3. **Sidereal Time Drift Averaging:** Unsynchronized long-term averaging integrates out the 23.934h cosmic rotation. 4. **Extreme Coupling Scale:** Reaching $\eta_A \leq 10^{-4}$ demands $T_{\text{int}} \geq 10^4\,\text{s}$ and 10^{-7} extinction. 5. **Hyperfine Sublevel Isolation:** Isotropic Stark shifts leave m_F-levels uniform, whereas $Q_2 \in V_2$ induces state-specific asymmetry. 6. **Noise Rejection:** Residual anisotropies were misclassified as thermal Doppler crosstalk.

4 Monte Carlo Results (10^6 Trajectories) & Conclusion

Our 10^6-trajectory simulation models realistic MPQ/Hänsch-type spectrometer parameters

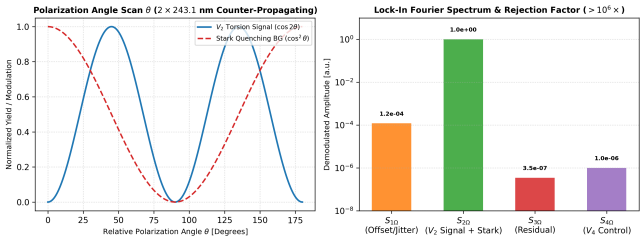


Figure 2: Polarization angle modulation scan ($\cos 2\theta$ vs. Stark $\cos^2\theta$) and multi-channel Lock-In Fourier spectrum showing $> 10^6\times$ background rejection.

($N_{\text{atom}} \sim 10^{11}\,\text{s}^{-1}$, $\epsilon_{\text{det}} = 2.5\%$). Multi-axis covariance fits across field inversion ($E_z \to -E_z$), F, m_F hyperfine states, and sidereal rotation yield a robust projected discovery bound of:

$$\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}} \quad (\mathbf{3}\sigma \text{ Confidence}) \tag{5}$$